Assignment 1

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Task 1: Repository Set-Up

Name	Last commit	Last update
bin	Build with Maven	50 seconds ago
cpp	finished move generation	21 hours ago
maps	Merge remote-tracking branch 'origin/main'	31 seconds ago
presentations	Small Changes	53 minutes ago
reports/Assignment1	Edit exercise4.tex	27 minutes ago
src/main/java	Build with Maven	50 seconds ago
target	Build with Maven	50 seconds ago
README.md	Initial commit	1 week ago
pom.xml	Build with Maven	50 seconds ago

Task 2: The Provided Toolset

```
Usage: ./server ( mapname | -s ) [ -b port number ] ( -t seconds | -d number ) [ -w number ] [ --matchpoint ] [ -r file | --moves file ] [ --gamestate ] [ --requestgamestate ] [ --observer ] [ --nocolor ]
 [ -v | -q ] [ -h ] [ --version ]

Arguments:
  mapname             Load this map
  -s, --standard      Load standard reversi map
  -b, --baseport      Use this port (NOTE: actual server port may differ if baseport is in use)
  -t, --timelimit     Apply timelimit (in seconds, >=0.2)
  -d, --depthlimit    Apply depthlimit (in case of time limit, clients may ignore this)
  -w, --wait          Seconds to wait for a single client to connect
  --matchpoint        Enable matchpoint mode (disable usual logging)
  -r, --replay        Specify moves to replay
  --moves             File to log moves to (for replay)
  --gamestate         Send the game state after every move
  --requestgamestate  Request the game state after every move
  --observer          Add an observer to the game

Basic options:
  --nocolor           Disable ANSI colors when printing
  -v, --verbose       Print more info
  -q, --quiet         Squelch output
  -h, --help          Show this help text
  --version           Show version information
```

Task 2: The Provided Toolset

```
Usage: ./client [ -s host or IP ] [ -p port number ] [ -g number ] [ --jitter ] [ --nocolor ] [ -v | -q ] [ -h ] [ --version ]

Arguments:
  -s, --server   Address of server to connect to
  -p, --port     Port to connect to
  -g, --group    Override the group ID
  --jitter       Print information about timing jitter

Basic options:
  --nocolor      Disable ANSI colors when printing
  -v, --verbose  Print more info
  -q, --quiet    Squelch output
  -h, --help     Show this help text
  --version      Show version information
```

Task 2: The Provided Toolset

```
Usage: ./ai [ -n ] [ --noprune ] [ --noaspiration ] [ -T | -S | -E ] [ -1 | -2 ] [ -w ] [ -s host or IP ] [ -p port number ] [ -g number ] [ --jitter ] [ --nocolor ] [ -v | -q ] [ -h ] [ --version ]

Arguments:
  -n, --nosort     Disable move-sorting
  --noprune        Disable pruning
  --noaspiration   Disable aspiration-windows
  -T, --trivial    Use trivial evaluation functions (default)
  -S, --simple     Use simple evaluation functions
  -E, --extended   Use extended evaluation functions
  -1, --paranoid   Use paranoid tree-searching (default)
  -2, --maxn       Use MAX^n tree-searching
  -w, --weaken     Weaken the AI (2/3 depth, 1/3 time)
  -s, --server     Address of server to connect to
  -p, --port       Port to connect to
  -g, --group      Override the group ID
  --jitter         Print information about timing jitter

Basic options:
  --nocolor        Disable ANSI colors when printing
  -v, --verbose    Print more info
  -q, --quiet      Squelch output
  -h, --help       Show this help text
  --version        Show version information
```

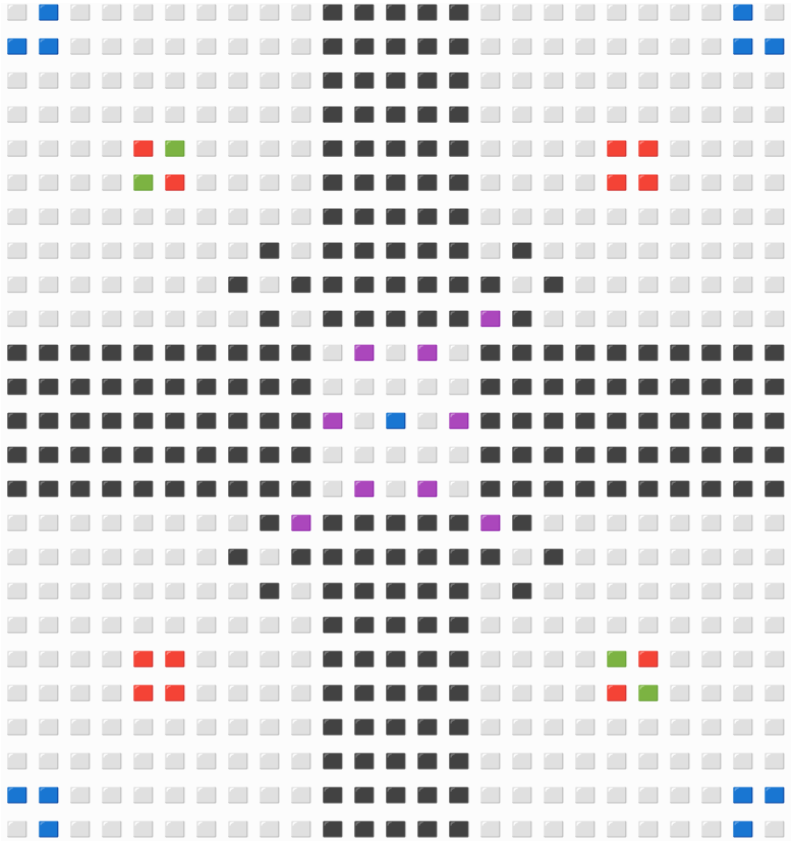
Task 2: The Provided Toolset

```
Usage: ./mapinfo filename... [ -t ] [ --nocolor ] [ -v | -q ] [ -h ] [ --version ]

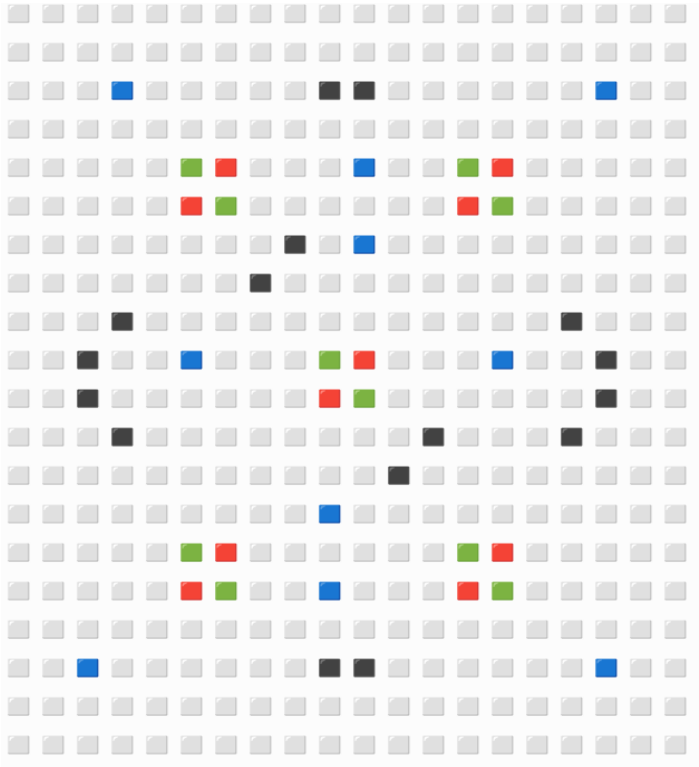
Arguments:
  filename       Analyze this map
  -t, --test     Only perform checks on the map, do not perform analysis

Basic options:
  --nocolor      Disable ANSI colors when printing
  -v, --verbose  Print more info
  -q, --quiet    Squelch output
  -h, --help     Show this help text
  --version      Show version information
```

Task 3: Your First Game Maps



For 4 players

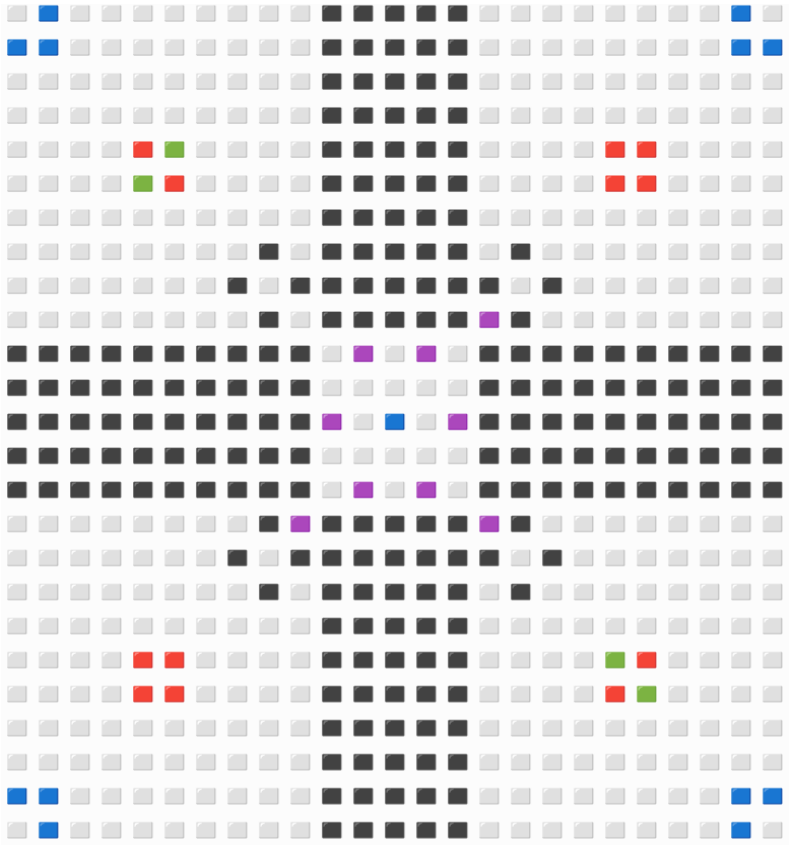


For 2 players

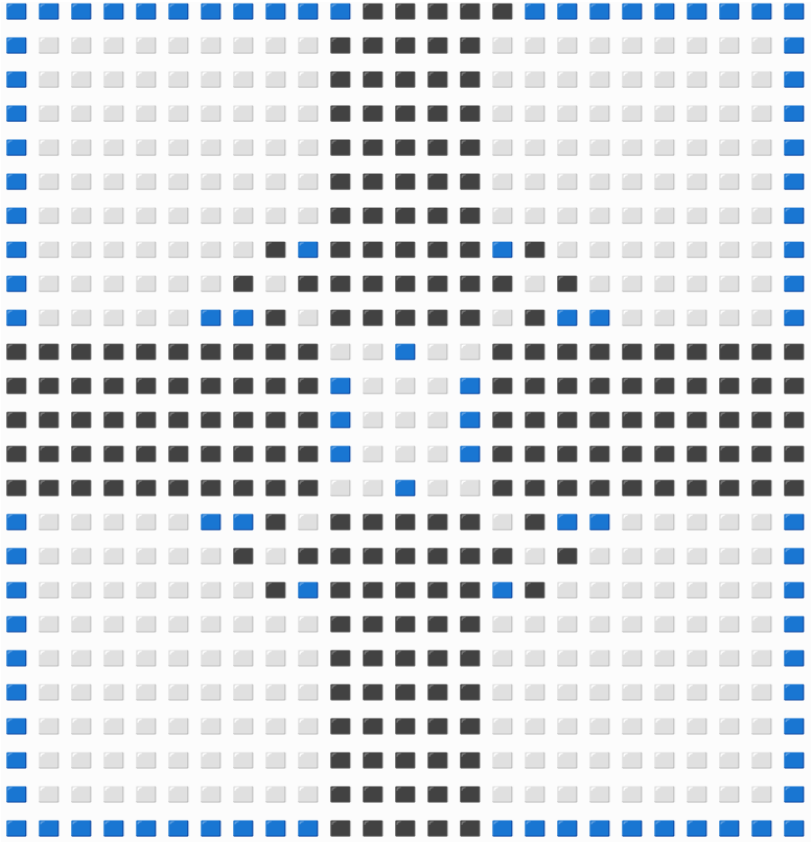
Task 3: Our Observations

- Holes: 'safe spaces', stopping bomb explosions
- Expansion stones: Rapid spreading (longer paths, 'Extension Rule')
- Bonus/choice/inversion: Balancing impact

Task 3: Exemplary Map



with player & element tiles



with transitions

Task 4: Map Data Structure

- Encode the board using multiple layers
- Each layer is responsible for representing one sort of tile
- Each sort of tile is represented by a layer except for empty tiles

Task 4: Map Data Structure

Micro Layer
(uint64_t)

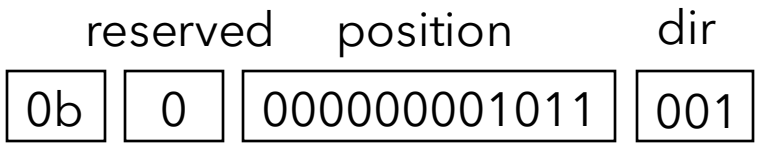
- N ≤ 64
- Less overhead
- Uses very efficient intrinsic functions for bit manipulation

Macro Layer
(std::bitset<2500>)

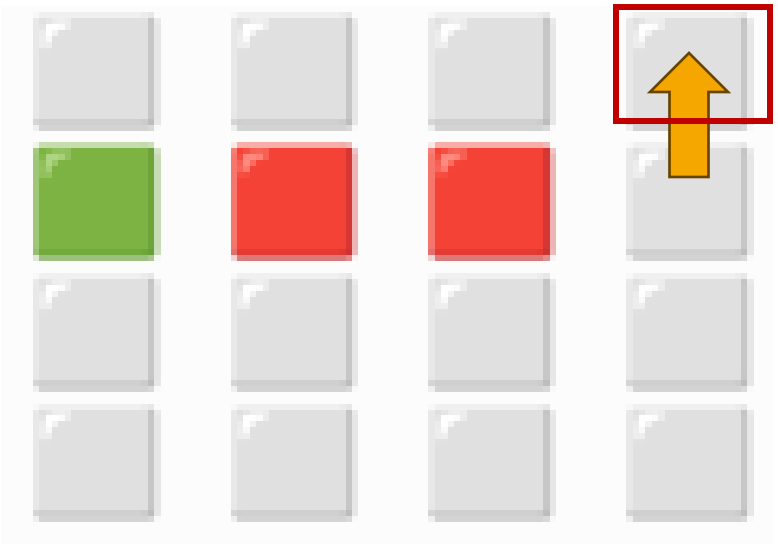
- N > 64
- Standard library, optimizied by compiler
- Fits in L1 Cache for N ≤ 2500

Task 4: Map Data Structure

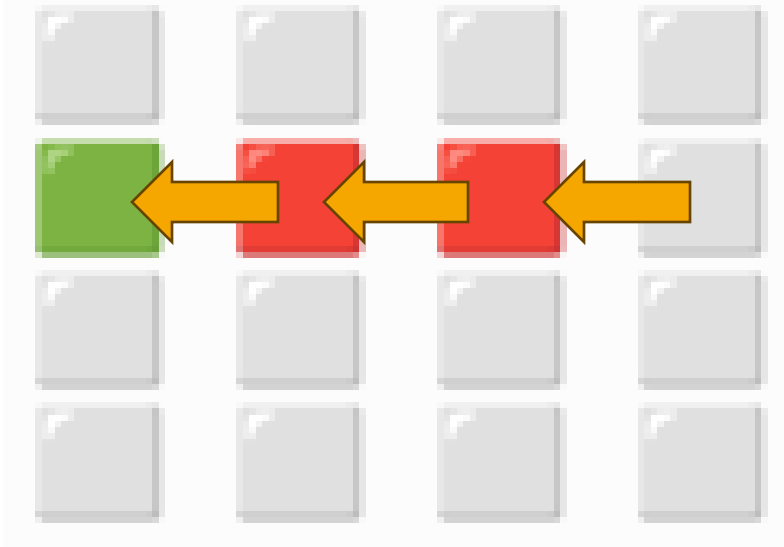
Transition domain encoding:



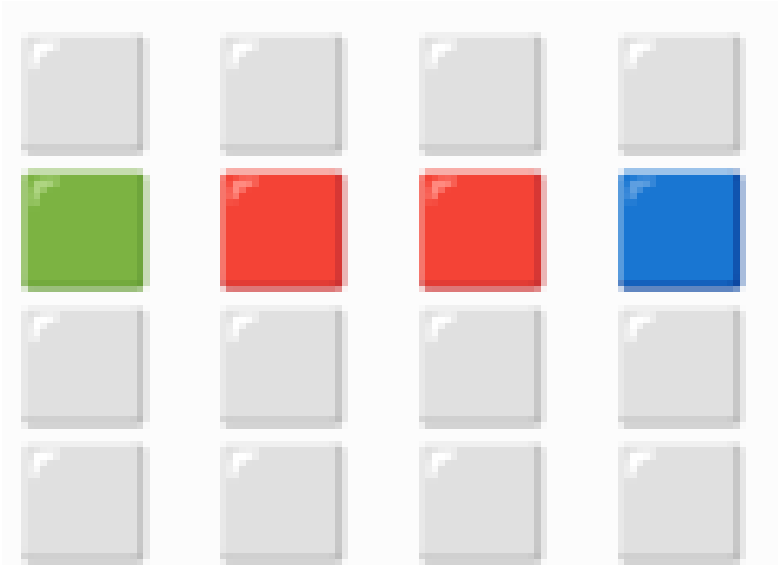
Task 5: Algorithm For Valid Moves



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Task 5: Algorithm For Valid Moves



Task 5: Algorithm For Valid Moves

- checkAllPlacable: find all valid moves. Iterate over all fields and save the ones where a stone could be placed
- isValidPlacing: use createsCapture to validare Override and normal moves
- createsCapture: check weather a placing whould result in a capture (DFS approach)

Task 5: Algorithm For Valid Moves

```
bool createsCapture(row, col):
    bool seenEnemy;
    // Durchlaufen für alle Himmelsrichtungen
    for each direction d:
        seenEnemy=false;
        newTransition = moveOneForward(row,col,d);
        // Durchlaufen, bis ggf. Maprand erreicht wird
        while ( inBounds() && isEnemyOrOverride(h, w)):
            newTransition = moveOneForward(row,col,d);
            seenEnemy = true;
        if (seenEnemy && inBounds && thisIsPlayer) return true;
    return false;
```

Task 6: Network Protocol Implementation

```
[ INFO] 21:47:01 Server is listening on port 7777
[ INFO] 21:47:01 Waiting for clients
[ INFO] 21:47:01 Waiting for player 1 to connect...
[ INFO] 21:47:33 Client connected
[ INFO] 21:47:33 Client is player 1
[ INFO] 21:47:33 Waiting for player 2 to connect...
[ INFO] 21:47:42 Client connected
[ INFO] 21:47:42 Client is player 2
[ INFO] 21:47:42 Waiting for group id of player 1
[ INFO] 21:47:42 OK, got a group: 7
[ INFO] 21:47:42 Waiting for group id of player 2
[ INFO] 21:47:42 OK, got a group: 255
[ INFO] 21:47:42 Client 0 connected as player 1 with group 7
[ INFO] 21:47:42 Client 1 connected as player 2 with group 255
[ INFO] 21:47:42 Sending data to player 1
[ INFO] 21:47:42 Sending data to player 2
[ INFO] 21:47:43 Init OK, starting game
[ INFO] 21:47:43 Timelimit 0
[ INFO] 21:47:43 Depthlimit 15
[ INFO] 21:47:43 Current map:
```

```
[ INFO] 21:47:43 Player 1 has 6 overrides and 4 bombs
[ INFO] 21:47:43 Sending move request to player 1 with depth limit: 15.
[ INFO] 21:47:43 Received move 1: Player 1 (x,y) = (7,5).
[ INFO] 21:47:43 Current board:
```

```
Connected to the server...
Sent group ID to the server: 7
Received message: Type=2, Length=464
Received map string.
Received message: Type=3, Length=1
Received player number: 1
Received message: Type=4, Length=5
Received move request.
Responded with move: (5, 7), special: 0
Received message: Type=6, Length=6
Another player made a move.
```